



# Assessing Argumentation Using Machine Learning and Cognitive Diagnostic Modeling

Xiaoming Zhai<sup>1,2</sup> · Kevin C. Haudek<sup>3,4</sup> · Wenchao Ma<sup>5</sup>

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## Abstract

In this study, we developed machine learning algorithms to automatically score students' written arguments and then applied the cognitive diagnostic modeling (CDM) approach to examine students' cognitive patterns of scientific argumentation. We abstracted three types of skills (i.e., attributes) critical for successful argumentation practice: making claims, using evidence, and providing warrants. We developed 19 constructed response items, with each item requiring multiple cognitive skills. We collected responses from 932 students in Grades 5 to 8 and developed machine learning algorithmic models to automatically score their responses. We then applied CDM to analyze their cognitive patterns. Results indicate that machine scoring achieved the average machine–human agreements of Cohen's  $\kappa = 0.73$ ,  $SD = 0.09$ . We found that students were clustered in 21 groups based on their argumentation performance, each revealing a different cognitive pattern. Within each group, students showed different abilities regarding making claims, using evidence, and providing warrants to justify how the evidence supports a claim. The 9 most frequent groups accounted for more than 70% of the students in the study. Our in-depth analysis of individual students suggests that students with the same total ability score might vary in the specific cognitive skills required to accomplish argumentation. This result illustrates the advantage of CDM in assessing the fine-grained cognition of students during argumentation practices in science and other scientific practices.

**Keywords** Cognitive diagnostic modeling · Argumentation · Machine learning · Constructed responses

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✉ Xiaoming Zhai  
Xiaoming.Zhai@uga.edu

<sup>1</sup> Department of Mathematics, Science, and Social Studies Education, University of Georgia, Athens, GA 30602, USA

<sup>2</sup> Institute for Artificial Intelligence, University of Georgia, Athens, GA 30602, USA

<sup>3</sup> CREATE for STEM Institute, Michigan State University, East Lansing, MI, 48824, USA

<sup>4</sup> Department of Biochemistry and Molecular Biology, Michigan State University, East Lansing, MI, 48824, USA

<sup>5</sup> College of Education, University of Alabama, Tuscaloosa, AL, USA

## Introduction

Argumentation is a scientific practice in which one coordinates evidence and theories to support or refute an explanatory conclusion, model, or prediction (Erduran et al., 2015). It is regarded as common sense that the natural world is filled with uncertainty (Lee et al., 2014); thus, one of the major endeavors of science is to generate knowledge and theories to eliminate uncertainty, explain phenomena, and uncover the underlying truth. To approach such a critical objective, scientists need to collect evidence to justify or critique claims. Osborne (2010) suggests that the most challenging arguments involve rebuttals or counter-arguments, as these arguments require one to not only collect evidence but also compare, contrast, and distinguish evidence and justify how the evidence warrants the claims. Though argumentation is critical, science education usually ignores this practice, due to the rush to present the major facts of science in the classroom. Consequently, students may perceive science as an authorized fact and lack the opportunity to practice the ability to critique—the latter being essential for creativity development (Henderson et al., 2015).

Educational research, including science curriculum standards, thus calls for engaging students in argumentation practice. Indeed, in the *Next Generation Science Standards* (National Research Council, 2012) argumentation is highlighted as one of eight critical practices in science education. Such an initiative can only be successful, however, if we have a robust understanding of students' cognitive competence in the argumentation (Fishman et al., 2017; Van den Eynde et al., 2019). Prior research has employed various approaches to examining how students utilize scientific knowledge, make claims, and reason when arguing (Kuhn & Reiser, 2005; Osborne et al., 2004; Simon et al., 2006), but limited knowledge exists regarding students' cognition in argumentation. We also have limited knowledge concerning whether students have cognitive patterns when making scientific arguments—and, if so, what these cognitive patterns might be.

Osborne et al. (2016) suggest that, due to the progression of learning and the variability of cognitive load required by argumentation tasks, students may demonstrate different levels of competence in argumentation when solving problems. The more proficient students are, the more likely they will be able to collect and use richer evidence to make valid claims. The authors employed a Rasch analysis approach to distinguish middle school students' performance levels with scientific argumentation; this analysis estimated students' ability based upon their performance on all the items the students attended. However, this approach may lack the nuance to gauge students' cognitive ability directly linked to the sub-dimensions of argumentation. A comprehensive review of the literature suggests that research is needed to assess students' cognitive patterns of argumentation based on the skills they need to complete the tasks (Erduran et al., 2015).

We therefore developed effective assessments to examine students' competence in argumentation, focusing on their cognitive patterns. By "effective assessments," we mean those that can elicit students' observable argumentation behaviors and help educators infer students' cognitive development regarding argumentation. Such assessments are usually performance-based in that they move away from traditional multiple-choice assessments, thus challenging teachers to grade students' responses and interpret the scores in a timely fashion (Lee et al., 2014). To bridge the assessment gap, we first developed constructed response tasks to assess argumentation. Our assessment tasks focused on eliciting three types of cognitive activities involved in argumentation: making claims, using evidence, and articulating why the evidence warrants the claims made. As noted earlier, the constructed responses move beyond the traditional multiple-choice items, and can elicit students'

knowledge-in-use (NGSS Lead States, 2018). To facilitate scoring efficiency, we developed machine learning algorithmic models to automatically score students' written responses (details about the development of automatic scoring can be found in Wilson et al., in press). Prior research suggests that algorithmic models that learn from human scoring can rapidly score students' written responses with accuracy near that of human experts (Zhai et al., 2020b, 2020c). However, it is not clear whether machine learning scoring models can score middle school student arguments with satisfactory accuracy for classroom use. Thus, in this study, we first examined the machine scoring accuracy of written arguments as part of a scientific argumentation instrument. Given our interest in diagnosing students' cognitive patterns of argumentation, we employed the cognitive diagnostic modeling (CDM) approach to analyze their cognitive patterns. These cognitive patterns are indicated by students' ability to make claims, use evidence, and articulate why the evidence warrants the claims made, with a focus on the complexity of the cognitive activities and intensity of the cognitive demands. Osborne et al. (2016) suggest that students' working memory limits their argumentation competence, and forms cognitive patterns as students learn to coordinate the core components of argumentation. These cognitive patterns may vary as students progress, since gaining familiarity with the phenomena and scientific knowledge is expected to reduce students' cognitive load regarding argumentation.

This study was therefore centered around two questions:

- (a) How accurate are the machine learning-based assessments for assessing argumentation?
- (b) What cognitive patterns do students exhibit in their argumentation practices?

## Assessing and Modeling Student Competence in Argumentation

### Argumentation in Science Education

Science education is rooted in the development of science (NRC, 2012), and one of the major drivers of science is argumentation. Indeed, the history of science is full of arguments (Osborne, 2010), and throughout the last century, arguments have played a significant role in engaging scientists in debate, driving the innovation of theories, and reforming ideas. For example, the well-known argument between Albert Einstein and Niels Bohr concerning how best to describe quantum mechanics illustrates how modern quantum theories were developed upon arguments (Einstein et al., 1935; Fine, 2004). It is generally accepted that the entire enterprise of science is grounded in arguments such as those comprising the Bohr–Einstein debates (Crombie, 1995; Osborne, 2010). Whether it is about proposing novel ideas or new theories, scientists need to justify the novelty and rationale of their ideas, convince the science community, and respond to critiques. Novel ideas are typically critiqued; this helps scientists improve the ideas and the underlying theories (Osborne, 2010). Moreover, the science community has developed the convention of peer review, during which reviewers evaluate and critique the claims, evidence, and/or warrants of the proposed arguments and conclusions in a study. Authors, in turn, must justify the reasonability of their arguments, refute non-grounded comments, or collect more evidence to enhance the validity of their conclusions. These activities constitute evidence for the importance of argumentation as a critical scientific practice to increase scientists' scientific literacy.

Asterhan and Schwarz (2016) suggest that, in students' argumentation practice in the science classroom, the primary goal should be to support the co-construction of

knowledge: i.e., *arguing to learn*. By engaging in argumentation, students use scientific knowledge to convince or persuade and justify their claims, which includes constructing an explanation (Nichols et al., 2015; Osborne & Patterson, 2011). Research also suggests that argumentation frequently appears in classroom discourse, where dialogue, social interaction, and learning environments play significant roles (McNeill & Krajcik, 2009; McNeill et al., 2016; Nichols et al., 2015; Zhu et al., 2017); this is not surprising, as research indicates that cognition is shaped by social interaction, during which verbal dialogue plays a critical role (Lemke, 1990). To capture the complexity of argumentation, Erduran et al. (2020) suggest that at least three theoretical bodies of literature can frame our understanding: developmental psychology, language sciences, and interdisciplinary investigations (e.g., history, philosophy, and sociology of science). These theoretical perspectives show how, by engaging in argumentation, students may demonstrate a range of cognitive abilities, in terms of collecting evidence or using evidence to justify, evaluate, or refute claims.

This study focused on the cognitive aspect of argumentation, as our assessments used written constructed responses, which involve less social and verbal interaction than discourse. We built upon Osborne et al. (2016) research on learning progression. Research shows that younger students or those with lower reasoning ability may perform more poorly in argumentation; for example, they may misuse personal observations as scientific evidence to justify claims or ignore evidence that contradicts their beliefs (Kuhn & Reiser, 2005). And although students' cognitive ability in argumentation may increase, it may not necessarily become mature. Indeed, as Liu and Roehrig (2017) found, teachers sometimes perform poorly on aspects of argumentation, particularly when justifying causal claims with evidence that only implies co-occurrence of events. In their work, Osborne et al. (2016) therefore assumed that students' argumentation competence would improve as they increasingly engaged in meaningful science learning. The authors verified a hypothesized learning progression of argumentation based on the cognitive load involved in the task. They adapted a framework proposed by Toulmin (1958), which included three components of arguments: claims, evidence, and warrants.

The relation between the evidence and the claim is provided by a warrant that forms the substance of the justification for the claim. Warrants may often be dependent on implicit assumptions that Toulmin referred to as backing. Claims may also be circumscribed by the use of qualifiers that define the limits of validity. (p. 825)

Osborne et al. (2016) argue that each argumentation task requires a given working memory, which accounts for students' varied levels of argumentation performance. Although Osborne et al. (2016) and other researchers (Lee et al., 2014, 2019; Mao et al., 2018) have contributed significantly to exploring students' argumentation performance, research is needed to further uncover the cognitive patterns exhibited by students when engaging in argumentation. Understanding students' cognition could help develop meaningful curriculum and instruction for varied levels of students.

## Assessing Scientific Argumentation

Assessment of argumentation is uniquely challenging, in two main ways: (a) eliciting the evidence and (b) modeling the measures. Eliciting the evidence is challenging in part because argumentation is a complex practice that usually involves interactions and communication (i.e., discourse) between idea holders (Henderson et al., 2018). While discourse may happen in classroom settings, it is infeasible to analyze students' arguments

and evaluate their competence in a timely fashion (Berland & McNeill, 2010; Bloom, 2001; Bravo & Chion, 2017; Duschl & Osborne, 2002; Hickey et al., 2012). An alternative approach to assessing argumentation is by asking students to write arguments regarding phenomena; written arguments are easily collected and nearly as effective as the discourse in soliciting students' thinking and performance on using evidence to support claims (Lee et al., 2014). More importantly, the development of natural language processing enables automatic and immediate scoring of students' performance, timely feedback, and automatic guidance (Lee et al., 2021; Linn et al., 2014; Zhu et al., 2017). This study thus employed written constructed responses to elicit students' argumentation performance.

Statistically modeling the measures of student argumentation performance is also challenging, but critical for assessment. Statistical modeling aims to assist educators in both identifying the patterns underlying scores and inferring students' competence. In scientific practices, the completion of each task often requires multiple cognitive skills. For example, to argue with others about phenomena, one has to collect and use evidence to support the claims one makes about the phenomena (McNeill & Krajcik, 2009). In real-life scenarios, such arguments can take place in multiple rounds, during which one could use various forms of evidence, make multiple claims, and infer from or refute others' claims. Such complex cognitive activities are challenging to model using classical testing theories, which lack the ability to tap the skill sets' nuances, given that students are usually scored holistically. In such settings, the scores do not necessarily reflect students' cognitive skills. To overcome this challenge, the current study employed CDM to model students' argumentation cognitive skills.

## A Cognitive Diagnostics Modeling Approach

CDMs are psychometric models for assessing students' mastery and non-mastery of a set of cognitive skills (e.g., scientific knowledge, use of evidence, making a claim) required to complete tasks in a test. While various types of CDMs have been developed, most of them need a Q-matrix (Tatsuoka, 1990), a binary item-by-skill association matrix, with 1 indicating that a skill is measured by an item and 0 indicating that a skill is not measured by the item. A Q-matrix is usually identified by experts with content knowledge who analyze which cognitive skills are needed to correctly solve a problem or conduct a task in the test (De La Torre & Minchen, 2014). Based upon students' test performance, psychometric models can be used to measure students' mastery patterns regarding the cognitive skills required by the test, providing teachers with diagnostic information. This measurement method is therefore termed "cognitive diagnostic modeling" or "cognitive diagnosis modeling." There are various methods with which to model students' cognitive skills. The current study used the linear logistic model (LLM) (Maris, 1999), a member of the general diagnostic model and one of the most widely used CDMs (Von Davier, 2008). We selected the LLM because the model assumes that each cognitive skill identified contributes to the probability of solving the problem, which better reflects the assessment tasks designed in this study.

Consider an item that measures two skills,  $\alpha_1$  and  $\alpha_2$ , each of which can have a value of 0 or 1 for a student, where 0 indicates that the skill is not mastered by the student and 1 indicates that the skill is mastered. For instance, a student with  $\alpha_1 = 1$  and  $\alpha_2 = 1$  masters both skills. Based on the LLM, the probability of answering this item correctly (i.e.,  $X = 1$ ) is defined as a nonlinear function of  $\alpha_1$  and  $\alpha_2$ , as in

$$\log \frac{P(X = 1)}{1 - P(X = 1)} = \delta_0 + \delta_1 \alpha_1 + \delta_2 \alpha_2$$

where  $\delta_0$  is the intercept with  $e^{\delta_0}/(1 + e^{\delta_0})$  being the probability of answering the item correctly for those who master neither skill and  $\delta_1$  and  $\delta_2$  are the increases in the log odds of success due to the mastery of skill 1 and skill 2, respectively. The ultimate goal of CDM analysis is to estimate the skill profile of each student based on their responses to the set of test questions. Using student profiles, we can classify students into different cognitive patterns and further study their cognitive abilities.

## Methods

### Participants

We recruited 932 students in science classes from 1 private school ( $n=128$ ) and 5 public middle schools ( $n=804$ ) in the Western United States. The students were distributed across Grades 5 to 8 ( $n_5=22$ ,  $n_6=166$ ,  $n_7=148$ ,  $n_8=514$ , undetermined=82). Of the sampled students, 14.2% were emergent English learners. Every student was administered the set of 19 items developed in this project.

### Instrument

To assess students' argumentation competence, we developed 19 assessment items in 3 groups. We began by employing student responses to an existing set of argumentation items developed in a previous project (Osborne et al., 2016) to train the machine algorithms. Learning from this process, we developed new items to improve machine prediction accuracy. Our team included experts in science education and assessment from three institutions who worked iteratively in the item development process to secure content validity (Wilson et al., under review). Each of the three groups of items shared the same context: group 1 shared the context of sugar dissolving in water (i.e., *sugar*); group 2 shared the same context in terms of gases (i.e., *gas*); and group 3 concerned the spread of bacteria (i.e., *bacteria*). Each item varied the cognitive demand aligned with a learning progression (Osborne et al., 2016). We also developed a rubric to score student performance on argumentation in terms of the claim made, use of evidence, and warrants of the inference using evidence to support the claim (for an example item and rubric, please refer to Appendixes 1 and 2).

### Automatic Scoring of Student Responses Using Machine Learning

Machine learning, an advanced artificial intelligence, denotes the technology by which computers (a) mimic humans' learning process to develop algorithmic models and then (b) apply the models to make decisions or predictions (Zhai, 2021). Machine learning has been increasingly applied in automatic scoring due to its powerful capacity to handle complex tasks (Nakamura et al., 2016; Zhai et al., 2021), including argumentation (Lee et al., 2021; Mao et al., 2018; Zhu et al., 2017). However, machine learning models are case specific—it is difficult to generalize a scoring model developed for one item to another, particularly for items focusing on different topics and with contextual features (Lintean et al., 2012). Therefore, the first part of the study examined the scoring accuracy of machine learning algorithmic models.

Developing a general machine learning-based scoring protocol involves two parts: training and testing. In the training process, human experts with content knowledge are hired to score student responses based on rubrics linked to students' proficiency. Scrutinized procedures are commonly employed to monitor and secure interrater reliability and solve discrepancies between human experts (for examples, see Maestrales et al. (2021)). Researchers then feed the computers the students' constructed responses and human scores to develop algorithmic models. According to a review study (Zhai et al., 2020b), a list of nearly 20 algorithms has been applied in science education to develop such algorithmic models. In a specific case, researchers may compare the algorithms, tune the parameters, and then eventually determine the optimal algorithm to develop the scoring models. However, most automatic scoring programs construct a test sample by splitting the fed data into a training and testing corpus. That is, the training data are used to develop the scoring model, which is applied to automatically score the testing corpus. The computer-assigned scores on the test samples are then compared to the human-assigned scores to examine the scoring accuracy. Certain standards must be met before the machine algorithmic models can be applied to score new data (Nehm et al., 2012).

In this study, we employed the Constructed Response Classifier (CRC) tool to develop the algorithmic scoring model for each item. The CRC employs an ensemble approach, which utilizes eight algorithms simultaneously to develop scoring models (Zhai et al., 2022; Zhai et al., 2020a). According to the models' performance, the computer assigned weights to the eight algorithms using a cross-validated weighted voting scheme (Large et al., 2019). This approach adopts and combines the advantages of individual algorithms and has shown greater scoring accuracy than individual algorithms (Jescovitch et al., 2020). To acquire the training data sets, we first had human experts manually score students' responses using rubrics developed for argumentation (for an example rubric, please see Appendix 2). During the training of the human experts, interrater reliability was assessed as higher than Cohen's  $k=.60$  before we processed the human scores to train the computer, although Cohen's  $k$  was usually much higher than .60 for many rubrics. For some items, we used three experts as scorers and thus calculated Krippendorff's alpha during human training. The interrater reliability for human experts across the three sets of items using Krippendorff's alpha was bacteria mean=.90; gases mean=.82; and sugar mean=.80 (Wilson et al., in press). Discrepancies in scores between experts were resolved by discussion between the experts until consensus was reached. According to Nehm and Haertig (2012) and Zhai et al. (2021), machine-human agreement ranging from .41 to .60 is moderate, .61 to .80 is substantial, and over .80 is almost perfect. We then utilized the human scores to train the machine and develop an algorithmic model for each assessment item. The number of responses in our training set is typical for the development of machine learning models for use at middle and secondary school levels (Zhai et al., 2021). To better examine the machine scoring accuracy, we employed a tenfold cross-validation approach, during which the computer first segmented the student responses with human scores into 10 even-number groups. Nine groups were used to train the machine and develop the scoring model; the model was then used to automatically score the leftover group. The machine-assigned scores were compared with human-assigned scores to calculate an accurate parameter of Cohen's  $k$ . The computer automatically rotated the 10 groups 10 times so that each group had an opportunity to serve as a testing data corpus, resulting in 10 Cohen's  $k$ . The average of the 10 Cohen's  $k$  indicates the accuracy of machine scoring.

## Cognitive Diagnostic Modeling Analysis

According to our theoretical framework, the set of items in the test measures three types of skills necessary for argumentation: *making claims (C)*, *using evidence (E)*, and *warranting (W)*.

Human experts with relevant content knowledge and knowledge about argumentation coded the items using the three types of skills and developed the Q-matrix according to Osborne et al. (2016) learning progression (see Table 7 in Appendix 3). For example, in order to correctly answer Question 2, students need to be able to make two claims, use one piece of evidence, and make two warrants. The LLM was fit to the data for estimating students' skill profiles. The monotonic constraints assumed that mastering an additional required attribute would not decrease the probability of success. This approach has been employed in prior research, which showed that it is intuitively reasonable and important for stabilizing the estimation of parameters (Ma & Jiang, 2021). Also note that skill hierarchy was specified based on the assumptions and theory. In particular, C1 is a prerequisite for C2 and C2 is a prerequisite for C3, or  $C1 \rightarrow C2 \rightarrow C3$ . This is because we assumed that C1 indicates that students were able to make one claim, while C2 indicates that students were able to make two claims, etc. The ability to make more claims suggests a higher cognitive ability according to the learning progression of the argumentation (Osborne et al., 2016). Similarly, we also have  $W1 \rightarrow W2 \rightarrow W3 \rightarrow W4 \rightarrow W5$  and  $E1 \rightarrow E2 \rightarrow E3$ .

To ensure that the inferences from CDMs are valid, we assessed the goodness-of-fit of the model to the data. In this study, we assessed the model–data fit by comparing the observed and the model-implied Pearson correlations between all item pairs and the observed and model-implied log odds between all item pairs (Chen et al., 2013). We also calculated the standardized root mean squared residual (SRMSR), which gauges the size of model–data misfit, and previous research suggests that a model with an SRMSR less than .05 can be viewed as a well-fitting model (Maydeu-Olivares & Joe, 2014). To assess the classification reliability, we calculated the attribute-level classification accuracy (Wang et al., 2015), defined as the percentage of agreement between the observed and expected proportions of examinees for each attribute. The LLM was estimated using the GDINA R package (Ma & de la Torre, 2020). The sample size is 932, large enough to obtain accurate parameter estimates according to Ma and Jiang (2021). Since the focus of the current CDM analysis is to estimate the skill mastery profiles of students in the current sample, the representativeness of the sample is less relevant.

## Findings

### Machine Scoring Accuracy Analysis

We first examined the machine–human scoring agreement to ensure that the machine scores were accurate enough to address our first research question (Table 1). The mean Cohen's  $\kappa = 0.73$ ,  $SD = 0.09$ . According to the criteria for the machine–human scoring agreement proposed by Nehm and Haertig (2012), Cohen's  $\kappa$  values between .41 to .60 are *moderate*, 0.61 to 0.80 is *substantial*, and 0.81 to 1.00 is (*almost*) *perfect*. In our study, the overall machine–human scoring agreement is substantial, with a number of individual models in the almost perfect range. Given that most items have more than two rubric levels, this range of machine–human scoring agreements is reliable.

### CDM Analysis of Argumentation

To examine model–data fit, we calculated the observed and the model-implied Fisher-transformed Pearson correlations and log odds ratios between all item pairs. The  $z$  statistics can be derived from the transformed correlations and log odds ratios, and hypothesis

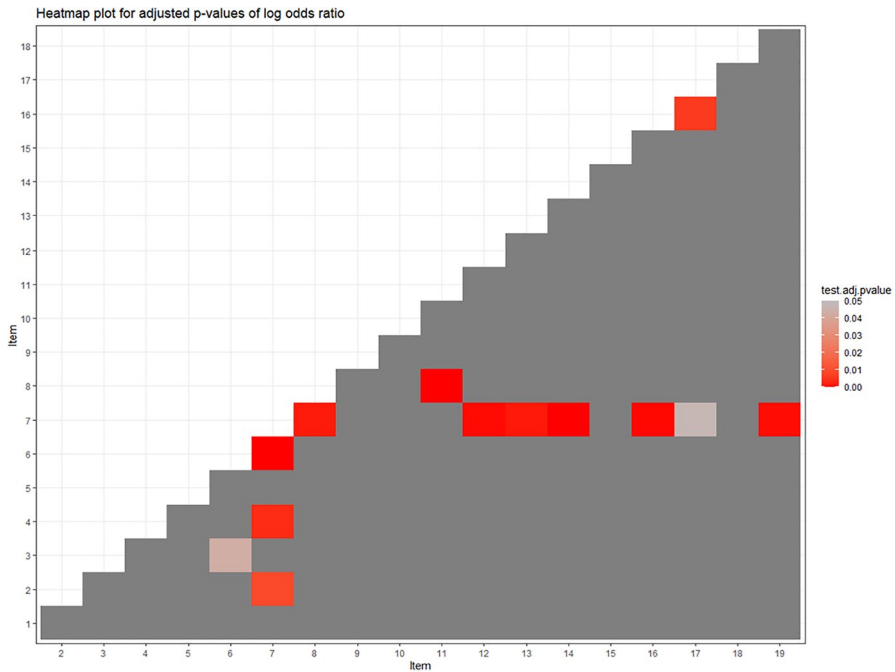
**Table 1** Machine–human scoring agreement for the 19 items

| Item | Machine–human agreement ( $\kappa$ ) | # of levels in rubric |
|------|--------------------------------------|-----------------------|
| 1    | 0.75                                 | 4                     |
| 2    | 0.82                                 | 2                     |
| 3    | 0.81                                 | 2                     |
| 4    | 0.73                                 | 4                     |
| 5    | 0.68                                 | 4                     |
| 6    | 0.93                                 | 3                     |
| 7    | 0.87                                 | 3                     |
| 8    | 0.69                                 | 4                     |
| 9    | 0.71                                 | 4                     |
| 10   | 0.74                                 | 4                     |
| 11   | 0.88                                 | 3                     |
| 12   | 0.63                                 | 2                     |
| 13   | 0.74                                 | 2                     |
| 14   | 0.60                                 | 4                     |
| 15   | 0.71                                 | 3                     |
| 16   | 0.71                                 | 3                     |
| 17   | 0.65                                 | 3                     |
| 18   | 0.59                                 | 3                     |
| 19   | 0.70                                 | 4                     |

tests can be performed to assess whether or not the discrepancies are significant. Note that when there are  $J$  items, there would be  $J*(J-1)/2$  transformed correlations and  $J*(J-1)/2$  log odds ratios, so the maximum  $z$  statistics were reported as test-level goodness-of-fit measures, and the corresponding  $p$  values were adjusted using the Holm–Bonferroni method for multiple comparisons. The SRMSR was also reported. Table 2 shows that the LLM cannot fit the data adequately (SRMSR = .06, adjusted  $p$  values were less than .01 for the maximum  $z$  scores) and the heatmap plot (Fig. 1), which displays the adjusted  $p$  values of log odds ratios between all item pairs, suggested that the LLM cannot explain the association between some item pairs related to items 7, 8, and 16. After the three items were removed, the LLM fit the data. As shown in Table 2, both transformed correlation and log odds ratios suggested that LLM can fit the data adequately at the .05 nominal level with an SRMSR = .04. This value is below the suggested threshold, allowing us to conclude that we have a well-fitting model. The following results are based on the analysis using the test without items 7, 8, and 16. Note that Item 7 is the only item that measures attribute E3. Therefore, after removing item 7, attribute E3 was removed from the Q-matrix.

**Table 2** Model comparison between 19 and 16 items

| Model    | SRMSR | Transformed correlation |                    | Log odds ratio |                    |
|----------|-------|-------------------------|--------------------|----------------|--------------------|
|          |       | max $z$ score           | adjusted $p$ value | max $z$ score  | adjusted $p$ value |
| 19 items | 0.06  | 6.49                    | <0.01              | 7.55           | <0.01              |
| 16 items | 0.04  | 3.49                    | 0.06               | 3.48           | 0.06               |



**Fig. 1** Heatmap plot of the items

Table 3 shows the probabilities of answering each item correctly for students who master all required skills for that item and for those who master none of those required attributes. The differences in the probability of success between these two groups of students—often referred to as a measure of item discrimination power—are also reported. It can be observed that most items performed quite well in that the discrimination indices were greater than 0.20. Two exceptions are items 1 and 11, which exhibited low discrimination below 0.10.

Table 4 shows the proportions of students who master each skill and the accuracy of skill estimation. It can be observed that the proportions of students who master each skill decreased along the hierarchy of the skills. For example, given that  $C1 \rightarrow C2 \rightarrow C3$ , the proportions of students who master C1, C2, and C3 were 0.82, 0.47, and 0.23, respectively. This pattern is true for all three types of skills—not unexpected in that skills nearer the end of the hierarchy tend to be harder. After CDM analysis, it is important to assess whether the classifications are accurate. Classification accuracy is a measure that concerns the extent to which the estimated attribute classifications and the true classifications are the same. The accuracies of the calculated skill classification were all greater than 0.7, suggesting that the students' skill estimates are in general accurate, according to Wang et al. (2015).

The CDM analysis showed that students could have distinct cognitive patterns of argumentation. CDM revealed 21 patterns, each with more than 1% of students (Table 5). For example, about 13.7% of students exhibited the mastery to propose three claims and use one piece of evidence connected with one warrant (class 1), while 7.5% of students did not master any of these argumentation skills (class 5).

For illustration purposes, Fig. 2 shows the probability of mastering each skill for three students from the data, each with a total score of 8 out of 16. The probability of mastery was calculated using estimated item parameters and the student's item responses and reflects our confidence in the

**Table 3** Probability of success by item

| Item | Probability of success |            | Discrimination |
|------|------------------------|------------|----------------|
|      | Master none            | Master all |                |
| 1    | 0.98                   | 1.00       | 0.01           |
| 2    | 0.00                   | 0.80       | 0.80           |
| 3    | 0.19                   | 0.66       | 0.47           |
| 4    | 0.55                   | 0.98       | 0.43           |
| 5    | 0.35                   | 1.00       | 0.65           |
| 6    | 0.30                   | 0.56       | 0.26           |
| 9    | 0.12                   | 1.00       | 0.88           |
| 10   | 0.76                   | 1.00       | 0.24           |
| 11   | 0.10                   | 0.18       | 0.08           |
| 12   | 0.51                   | 0.95       | 0.43           |
| 13   | 0.44                   | 0.86       | 0.43           |
| 14   | 0.21                   | 0.83       | 0.63           |
| 15   | 0.15                   | 0.70       | 0.55           |
| 17   | 0.06                   | 0.58       | 0.52           |
| 18   | 0.08                   | 0.38       | 0.29           |
| 19   | 0.00                   | 1.00       | 1.00           |

classification of the student. It can be observed that they exhibited distinct mastery probabilities for these three types of skills. For example, Student A had very high probabilities of mastering C1, E1, and E2, but very low probabilities of mastering C2, C3, W2, W3, W4, and W5, whereas Student B had high probabilities of mastering E1 and W1, but low probabilities of mastering other skills. Student C had a different profile from both Students A and B in that they had high probabilities of mastering C1 and W1, medium probabilities of mastering other warrants, but very low probabilities of mastering evidence skills. This suggests that the total item score is too coarse-grained to be useful in providing detailed diagnostic information about students' mastery profiles.

## Discussion and Conclusions

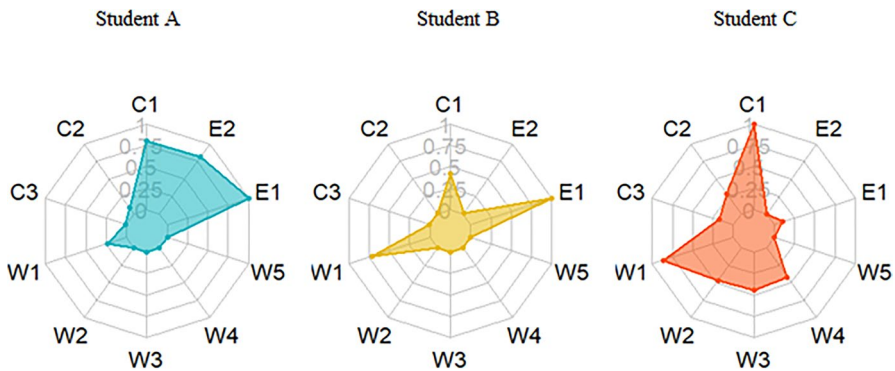
This study contributes to our understanding of middle school students' cognitive patterns of argumentation. We found that students were clustered in 21 groups based upon their argumentation performances, each revealing a different cognitive pattern. Within each group, students showed different abilities regarding proposing claims, using evidence, and providing warrants to justify how the evidence supports a claim. The nine most frequent groups accounted for more than 70% of the students in the study. These findings suggest that argumentation is a complex cognitive process and that many students are facing challenges.

**Table 4** Accuracy of estimated attributes

|            | C1   | C2   | C3   | W1   | W2   | W3   | W4   | W5   | E1   | E2   |
|------------|------|------|------|------|------|------|------|------|------|------|
| Proportion | 0.82 | 0.47 | 0.23 | 0.62 | 0.19 | 0.15 | 0.13 | 0.04 | 0.56 | 0.14 |
| Accuracy   | 0.89 | 0.72 | 0.74 | 0.84 | 0.87 | 0.87 | 0.88 | 0.96 | 0.84 | 0.88 |

**Table 5** Argumentation mastery patterns revealed from CDM analysis

| C1 | C2 | C3 | W1 | W2 | W3 | W4 | W5 | E1 | E2 | %     |
|----|----|----|----|----|----|----|----|----|----|-------|
| x  | x  | x  | x  |    |    |    |    | x  |    | 13.7% |
| x  |    |    | x  |    |    |    |    |    |    | 9.3%  |
| x  |    |    | x  | x  | x  | x  |    |    |    | 8.6%  |
| x  | x  |    |    |    |    |    |    | x  |    | 7.8%  |
|    |    |    |    |    |    |    |    |    |    | 7.5%  |
|    |    |    |    |    |    |    |    | x  |    | 7.0%  |
| x  | x  | x  |    |    |    |    |    |    |    | 6.5%  |
| x  | x  |    | x  |    |    |    |    | x  | x  | 5.3%  |
| x  |    |    | x  |    |    |    |    | x  |    | 5.2%  |
| x  |    |    | x  |    |    |    |    | x  | x  | 3.7%  |
| x  | x  |    | x  | x  |    |    |    |    |    | 3.4%  |
| x  |    |    |    |    |    |    |    | x  | x  | 3.1%  |
| x  | x  |    | x  |    |    |    |    | x  |    | 2.5%  |
| x  | x  |    |    |    |    |    |    |    |    | 2.3%  |
| x  |    |    | x  | x  | x  | x  | x  |    |    | 2.1%  |
| x  | x  | x  |    |    |    |    |    | x  |    | 1.7%  |
|    |    |    | x  |    |    |    |    | x  |    | 1.6%  |
| x  | x  |    | x  |    |    |    |    |    |    | 1.5%  |
| x  |    |    |    |    |    |    |    |    |    | 1.2%  |
|    |    |    |    |    |    |    |    | x  | x  | 1.2%  |
| x  | x  | x  | x  | x  | x  |    |    | x  |    | 1.0%  |

**Fig. 2** Probability of mastering each skill for three students with the same total score

This study examined the accuracy of machine learning-based scoring for assessing argumentation. Overall, we successfully developed 19 machine learning algorithmic models that showed a high degree of agreement between the machine- and human-assigned scores. All models except one showed substantial to (almost) perfect agreement with human scores. This is encouraging, given that most models were for items that had > 2 scoring levels. When examining the individual assessment items, we found only two items that had low discrimination measures. In one case, this was the result of non-mastery students

performing well on the item, and in the other case, it was due to mastery students not being able to successfully respond to the item. All other items used in this study allowed good discrimination between students who mastered the required attributes and students who mastered none of the attributes. Therefore, we were able to use machine algorithms to score a set of argumentation items to examine patterns of student mastery of argumentation. Aligned with prior research (Mao et al., 2018), this finding shows the potential of machine learning in automatic scoring of student-written arguments; in their study, Mao et al. developed machine learning algorithms using c-rater-ML to automatically score student argumentations and achieved satisfactory accuracy. The c-rater-ML employed one single support vector regression (SVR) algorithm, while we employed the CRC, which assembles eight algorithms. The eight algorithms worked simultaneously and received weight in the final prediction according to their performance. Research has compared the individual algorithm and the ensemble approach and found the latter to be more accurate in most cases (Jescovitch et al., 2020). While the assessment tasks used by Mao et al. (2018) explicitly asked students to explain their uncertainty when explaining a phenomenon, our assessment tasks focused on eliciting students' ability to convince or refute others through evidence-based justification. This shift in the argumentation focus is reflected in both our assessment and rubrics, as well as the cognitive load required for students to complete the task. Moreover, our assessment tasks targeted different levels of cognitive load aligned with a learning progression (Osborne et al., 2016). The design approach we employed potentially involved a complex and diverse construct (Haudek & Zhai, 2021), which may have created an additional challenge for automatic scoring.

We then employed CDM to uncover patterns in students' argumentation practice. Prior research (Osborne et al., 2016) has explicitly considered the cognitive load in designing tasks to examine students' argumentation learning progression. However, due to the statistical methods employed that holistically estimate students' cognitive ability, it has been challenging to investigate the nuance of students' cognitive skills. This study therefore employed CDM, a measurement method that can gauge students' cognitive patterns, to assess students' argumentation practices. CDM uncovered not only students' cognitive skills in terms of making claims, using evidence, and providing warrants to justify how the evidence supports the claims, but also provided diagnostic information about the diversity of students' cognitive patterns. Encouragingly, we found that nearly all students (i.e., 92%) demonstrated mastery of at least one argumentation attribute. Most mastered one claim attribute, and nearly half mastered a second claim attribute in their arguments. It should be noted that making two claims for one phenomenon was significantly more cognitively demanding than double the cognitive demands of making an individual claim. This is because the claims made were based upon deep reasoning and scientific knowledge that students could apply to explain the phenomenon (Asterhan & Schwarz, 2016). We also found that most students had mastered one warrant attribute and one piece of evidence as part of their argumentation practice. However, in contrast to making claims, few students had mastered additional warrants or evidence attributes. Overall, most students mastered the first-level attribute for all claims, warrants, and evidence; they also demonstrated mastery of a second-level attribute for at least one claim, warrant, and evidence. This suggests that students could be successful in incorporating multiple elements of argumentation into their arguments, which is the foundation for building more complex arguments (e.g., compound or two-sided arguments) (Schwarz et al., 2003).

CDM also showed the mastery patterns of cognitive skills that were most frequent among students. We identified a total of 21 patterns, with the most common mastery pattern exhibited by about 14% of students. This large number of patterns is unsurprising given that the students

were distributed between Grades 5 and 8, across 5 schools. These students had mastered providing up to three claims and one piece of evidence and using one warrant. Interestingly, the third most frequent cluster, exhibited by ~9% of students, demonstrated that students could use multiple warrants during argumentation practice. This is important since using multiple warrants (reasons) when engaged in argumentation has been found to be associated with fewer one-sided arguments (Schwarz et al., 2003). Items requiring students to master multiple warrant attributes were usually highly complex and targeted higher levels of learning progression (Osborne et al., 2016). There were nine frequent patterns, each of which was exhibited by 5% or more of the students. Of these nine, only one cluster showed students with mastery of more than one warrant. Similarly, only one cluster showed that students had mastery of providing multiple pieces of evidence. Taken with the findings discussed above, this suggests that while students were able to master second- and third-level attributes, students showed high variability concerning the elements (e.g., claim, warrant, evidence) these higher-level attributes fit. This is exemplified in Fig. 2, which shows probable mastery patterns for three students with identical overall scores on the instrument. Here, we realized that two of the three students had a high probability of demonstrating mastery of second-level attributes. However, for Student A, the second-level attribute was associated with evidence, while for Student C, the most likely higher-level attributes were for warrants.

These findings have significant implications for how we (a) view student learning of argumentation practice and (b) provide instructional support. To our knowledge, this is the first application of CDM to assess students' proficiency in scientific argumentation; here, CDM is shown to be a valuable tool for revealing how students succeed and need support in this critical scientific practice. As part of this study, we used an existing learning progression for scientific argumentation. This progression describes three broad levels of student proficiency, with several sub-levels (Osborne et al., 2016). As noted by Osborne et al., the sub-levels are not ordered, and therefore some possible resolution is not available when considering student performance at a broad level. Our CDM results offer an alternative perspective in that they reveal which "finer-grained" argumentation attributes are most likely to be mastered by students. As might be predicted from the learning progression, not all students at a broad learning progression level are homogenous in their proficiency in an identical set of argumentation skills. Furthermore, the mastery patterns identified by CDM may help us understand different ways students learn to master the practice of argumentation. This would require following students over time to examine how they move from one cluster to another; how the mastery of some attributes is associated with transitioning between learning progression levels; and how/when the cluster pattern changes as argumentation becomes more complex. That is, although learning progressions are based on developmental milestones (Corcoran et al., 2009), students may exhibit different paths in achieving those milestones (Gao et al., 2020, 2021); techniques such as CDM may help reveal patterns among these paths.

Automated scoring systems offer opportunities to provide (near) real-time feedback to students when responding to assessment tasks, including scientific argumentation (Cavalcanti et al., 2021; Zhai et al., 2020b). Prior research has shown that students' arguments improved in quality when such systems were utilized (Zhu et al., 2017). However, our CDM findings suggest that not all students require the same support and that using information from sets of items to assign students to groups or patterns may enable the provision of more personalized feedback and/or follow-up tutorials or assessments. For example, instructional support for Student A (from Fig. 2) should focus on the mastery of warrants, while support for Student B should focus on the mastery of second-level attributes across all elements. Therefore, incorporating common patterns of student argumentation into automated systems as part of task- and process-level feedback may support a greater number of students to develop more complex argumentation skills (Hattie & Timperley, 2016).

## Limitations

Although this study presents interesting findings, there are limitations. First, we found that some items were more challenging to automatically and accurately score than others, but we did not examine the accountability. We also found that for some items, even human experts encountered difficulty in achieving robust interrater reliability. Moreover, it is not clear whether the machine–human agreements were causally associated with human–human agreements. Future studies should employ qualitative methods to further investigate these problems.

## Appendix 1. Example item

**Bacteria item.** A student at school says that automatic hand dryers in public bathrooms are dangerous. This student shows you a picture of bacteria growing on a petri dish that was placed under an automatic hand dryer in a public bathroom for 3 min. In the picture, the petri dish is covered with several types of bacteria.



### Key for Picture

#### Types of Bacteria:

Type 1 

Type 2 

Type 3 

Type 4 

Type 5 

Petri Dish 

IPOPBA / GETTY (LEFT) NICHOLE WARD / FACEBOOK (RIGHT)

1. (Level 0b) What claim is the student making?

## Appendix 2. Example rubric

**Table 6**

**Table 6** Example rubric for Question 1 of Bacteria item

| Component                                                                                                          | 0/1    | Examples (include I.D. #)                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Component A: Claim                                                                                                 |        |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| A1: Hand dryers (or suitable pronoun reference)                                                                    | 1 or 0 | 8. That automatic <i>hand dryers</i> in public bathrooms are a health hazard<br>27. Automatic <i>hand dryers</i> are dangerous<br>27. Automatic hand dryers <i>are dangerous</i>                                                                                                                                                                                                                                              |
| A2: are dangerous (or synonyms of dangerous including “unsafe” or “bad” or “unsanitary” but NOT including “dirty”) | 1 or 0 |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Component B: Evidence                                                                                              |        |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| B1: Includes any of the following: Bacteria                                                                        |        | The student is claiming that automatic hand dryers are dangerous <i>due bacteria spread</i> by the use of them<br>The student is claiming that dryers might be dangerous, and <i>the bacteria</i> came from the automatic dryer                                                                                                                                                                                               |
| B2: Petri dish                                                                                                     |        |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| B3: Several types (or “kinds” but NOT including “lots”)                                                            |        |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Holistic score                                                                                                     |        |                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Both claim components present and clearly identified BUT NO Evidence components                                    | 2      | 3. Automatic <i>hand dryers</i> in public bathrooms <i>are dangerous</i><br>5. Automatic <i>hand dryers</i> in public bathrooms <i>are dangerous</i><br>8. That automatic <i>hand dryers</i> in public bathrooms <i>are a health hazard</i><br>27. Automatic <i>hand dryers</i> <i>are dangerous</i><br>43. That automatic <i>hand dryers</i> in public bathrooms <i>are dangerous</i><br><i>Are dangerous OR hand dryers</i> |
| Either claim component present BUT NO Evidence components                                                          | 1      | 2. The student is claiming that automatic <i>hand dryers</i> <i>are dangerous due bacteria</i> spread by the use of them                                                                                                                                                                                                                                                                                                      |
| Any of the evidence components are present                                                                         | 0      | 6. The student is saying that <i>automatic hand dryers are dangerous</i> because they have a lot of <i>bacteria</i> growing on them<br>35. That automatic <i>hand dryers are dangerous</i> and said that <i>bacteria</i> grew under it<br>38. Those public bathrooms <i>are dangerous</i> when there are <i>hand dryers</i> in them <i>due to the bacteria</i> that comes from them                                           |
| Absence of any components (claim or evidence)                                                                      | 0      |                                                                                                                                                                                                                                                                                                                                                                                                                               |

## Appendix 3. Data matrix

**Table 7**

**Table 7** Q matrix of the items and the attributes

| Item | C1 | C2 | C3 | W1 | W2 | W3 | W4 | W5 | E1 | E2 | E3 |
|------|----|----|----|----|----|----|----|----|----|----|----|
| 1    | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 0  | 0  |
| 2    | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  |
| 3    | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  |
| 4    | 1  | 1  | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 0  | 0  |
| 5    | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0  | 1  | 1  | 0  |
| 6    | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 7    | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 1  |
| 8    | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 0  | 0  |
| 9    | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0  |
| 10   | 1  | 1  | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 0  | 0  |
| 11   | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 12   | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 13   | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 14   | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 15   | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 16   | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 17   | 1  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 1  | 0  | 0  |
| 18   | 1  | 1  | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 0  | 0  |
| 19   | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 0  | 1  | 1  | 0  |

C, claim; W, warrant; E, evidence

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**Data Availability** Data is available by contacting the corresponding author.

## Declarations

**Ethics Approval and Consent to Participate** All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. This article refers to no studies involving animals conducted by any of the authors. Informed consent was obtained from all individual participants included in the study.

**Consent for Publication** All authors agreed to publish the study in this journal.

**Competing Interests** The authors declare no competing interests.

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